

CS57300
PURDUE UNIVERSITY

DATA MINING

SEARCH

MAXIMUM LIKELIHOOD ESTIMATION

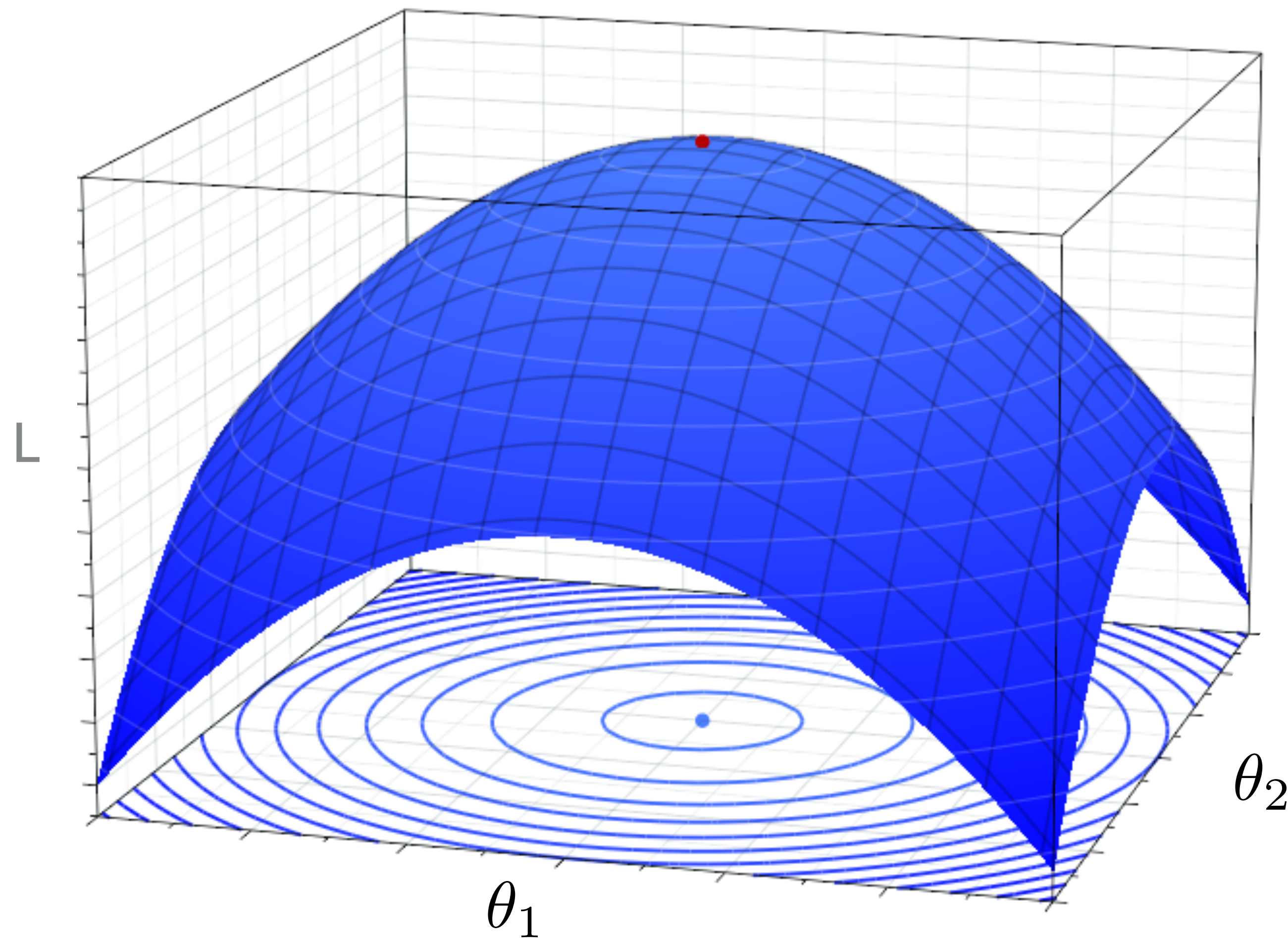
- ▶ “Learn” the best parameters by finding the values of θ that maximizes likelihood:

$$\hat{\theta}_{MLE} = \arg \max_{\theta} L(\theta)$$

- ▶ Often easier to work with loglikelihood:

$$\begin{aligned} l(\theta|D) &= \log L(\theta|D) \\ &= \log \prod_{i=1}^n p(x(i)|\theta) \\ &= \sum_{i=1}^n \log p(x(i)|\theta) \end{aligned}$$

LIKELIHOOD SURFACE



If the likelihood surface is convex we can often determine the parameters that maximize the function analytically

MLE FOR NBC

► Likelihood:
$$L(\theta | D) = \prod_{i=1}^n \prod_{j=1}^m P(x_{ij} | c_i) P(c_i)$$

MLE FOR NBC

- ▶ Likelihood: $L(\theta | D) = \prod_{i=1}^n (\prod_{j=1}^m P(x_{ij} | c_i)) P(c_i)$
- ▶ Parameters:
 - ▶ Priors: Denote $p_l = P(c_i = l)$, $l = 1, \dots, L$ (i.e., there are L possible labels)
 - ▶ Conditional probability distributions: Denote $q_l^{jk} = P(x_{ij} = k | c_i = l)$, $k = 1, \dots, K(j)$ (i.e., the j -th attribute has $K(j)$ possible value)

MLE FOR NBC

- ▶ Rewrite likelihood: $L(\theta|D) = \left(\prod_{l=1}^L p_l^{N_l}\right) \left(\prod_{l=1}^L \prod_{j=1}^m \prod_{k=1}^{K(j)} (q_l^{jk})^{N_l^{jk}}\right)$
 - ▶ $N_l = \sum_{i=1}^n I(c_i = l)$, i.e., the number of data points in class l
 - ▶ $N_l^{jk} = \sum_{i=1}^n I(c_i = l, x_{ij} = k)$, i.e. the number of data points in class l , and its j -th attribute is k

MLE FOR NBC

- ▶ Rewrite likelihood: $L(\theta|D) = \left(\prod_{l=1}^L p_l^{N_l}\right) \left(\prod_{l=1}^L \prod_{j=1}^m \prod_{k=1}^{K(j)} (q_l^{jk})^{N_l^{jk}}\right)$
 - ▶ $N_l = \sum_{i=1}^n I(c_i = l)$, i.e., the number of data points in class l
 - ▶ $N_l^{jk} = \sum_{i=1}^n I(c_i = l, x_{ij} = k)$, i.e. the number of data points in class l , and its j -th attribute is k
- ▶ Convex maximization
 - ▶ $p_l = N_l/n$, i.e., the fraction of data in the training set where its label is l
 - ▶ $q_l^{jk} = N_l^{jk}/N_l$, i.e. the fraction of data whose j -th attribute is k among data whose label is l

LEARNING CPDS FROM EXAMPLES

		X_1		
		Low	Med	High
Y	Yes	10	13	17
	No	2	13	0

$$P[X_1 = \text{Low} \mid Y = \text{Yes}] = \frac{10}{(10 + 13 + 17)}$$

$$P[Y = \text{No}] = \frac{(2 + 13)}{(2 + 13 + 10 + 13 + 17)}$$

NBC LEARNING

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

- Estimate prior $P(BC)$ and conditional probability distributions $P(A \mid BC)$, $P(I \mid BC)$, $P(S \mid BC)$, $P(CR \mid BC)$ independently with maximum likelihood estimation

P(BC)

BC	θ
yes	9/14
no	5/14

P(A | BC)

BC	A	θ
yes	<= 30	2/9
	31..40	4/9
	> 40	3/9
no	<= 30	3/5
	31..40	0/5
	> 40	2/5

P(I | BC)

BC	I	θ
yes	high	2/9
	med	4/9
	low	3/9
no	high	2/5
	med	2/5
	low	1/5

P(S | BC)

BC	S	θ
yes	yes	6/9
	no	3/9
no	yes	1/5
	no	4/5

P(CR | BC)

BC	CR	θ
yes	exc	3/9
	fair	6/9
no	exc	4/5
	fair	1/5

NBC PREDICTION

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no
31..40	high	no	excellent	?

► What is the probability that new person will buy a computer?

$$P(BC = yes|A = 31..40, I = high, S = no, CR = exc)$$
$$\propto P(A = 31..40|BC = yes)P(I = high|BC = yes)$$
$$P(S = no|BC = yes)P(CR = exc|BC = yes)P(BC = yes)$$

P(BC)

BC	θ
yes	9/14
no	5/14

P(A | BC)

BC	A	θ
	<= 30	2/9
yes	31..40	4/9
	> 40	3/9
no	<= 30	3/5
	31..40	0/5
	> 40	2/5

P(I | BC)

BC	I	θ
	high	2/9
yes	med	4/9
	low	3/9
no	high	2/5
	med	2/5
	low	1/5

P(S | BC)

BC	S	θ
yes	yes	6/9
	no	3/9
no	yes	1/5
	no	4/5

P(CR | BC)

BC	CR	θ
yes	exc	3/9
	fair	6/9
no	exc	4/5
	fair	1/5

IS THERE ANY PROBLEM?

		X_1		
Y		Low	Med	High
	Yes	10	13	17
	No	2	13	0

ZERO COUNTS ARE A PROBLEM

- ▶ If an attribute value does not occur in training data, we assign **zero** probability to that value
- ▶ How does that affect the conditional probability $P[f(x) | x]$?
- ▶ It equals 0!!!
- ▶ Why is this a problem?
- ▶ Adjust for zero counts by “smoothing” probability estimates

SMOOTHING: LAPLACE CORRECTION

		X_1		
		Low	Med	High
Y	Yes	10	13	17
	No	2	13	0

Laplace correction

Numerator: **add 1**

Denominator: **add k** ,
where k =number of
possible values of X

$$P[X_1 = \text{High} \mid Y = \text{No}] = \frac{0}{(2 + 13 + 0)} \frac{+1}{+3} \text{ Adds uniform prior}$$

WHAT ABOUT CONTINUOUS VARIABLES

- ▶ Discretize continuous variables through binning
 - ▶ Split the range of the continuous variable to several bins, assign a categorical value to each bin, and map continuous values fall into that bin to the assigned categorical value
- ▶ Model the probability distribution for continuous variables explicitly
 - ▶ For example, assume a Gaussian distribution and introduce additional parameters: $P(x_{ij} = x | c_i = l) \sim N(\mu_j^l, \sigma_j^l)$

IS ASSUMING INDEPENDENCE A PROBLEM?

- ▶ What is the effect on probability estimates?
 - ▶ Over-counting evidence, leads to overly confident probability estimate
- ▶ What is the effect on classification?
 - ▶ Less clear...
 - ▶ For a given input x , suppose $f(x) = \text{True}$
 - ▶ Naïve Bayes will correctly classify if $P[f(x) = \text{True} \mid x] > 0.5$
...thus it may not matter if probabilities are overestimated